

# Workshop: Differentiation Product and Quotient Rules

UEA LEARNING ENHANCEMENT TEAM MATHEMATICS & STATISTICS

## Product Rule

Functions that are made of other functions multiplied together can be differentiated using the product rule.

$$y = f(x) \times g(x)$$

For example, you may have the function  $y = x^2 e^x$ .

Use 'dummy' variables to represent the two functions:

$$u = f(x), \quad v = g(x)$$

Then:

$$\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$$

So, in the example:

$$\begin{aligned} u &= x^2, & v &= e^x \\ \frac{du}{dx} &= 2x, & \frac{dv}{dx} &= e^x \\ \frac{dy}{dx} &= 2xe^x + x^2 e^x = xe^x(2 + x) \end{aligned}$$

Differentiate the following:

1.  $y = 5xe^x$
2.  $y = x^3 \ln x$
3.  $y = 4e^x \ln 4x$
4.  $y = (6x^2 + 4x + 1) \ln x^2$
5.  $y = \left(5x - \frac{3}{x^2}\right) e^x$

## Quotient Rule

When a function is produced by a function divided by another function, it can be differentiated with the quotient rule.

$$y = \frac{f(x)}{g(x)}$$

$$\text{Let } u = f(x) \text{ and } v = g(x)$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

For example:  $y = \frac{x^2+5}{2x-1}$

$$u = x^2 + 5, \quad v = 2x - 1$$

$$\frac{du}{dx} = 2x, \quad \frac{dv}{dx} = 2$$

$$\frac{dy}{dx} = \frac{(2x-1) \times 2x - (x^2+5) \times 2}{(2x-1)^2} = \frac{4x^2 - 2x - 2x^2 + 10}{(2x-1)^2}$$

$$\frac{dy}{dx} = \frac{2x^2 - 2x + 10}{(2x-1)^2}$$

Differentiate the following:

6.  $y = \frac{5x+2}{x-4}$

7.  $y = \frac{e^x}{x^2+2x}$

8.  $y = \frac{\ln 4x}{x^2+5}$

9.  $y = \frac{\sqrt{x}}{e^x}$

### Mixed

The following questions use product and quotient rule. They also make use of the chain rule. Find the derivative

$$10. y = e^{4x}(5x + 2)$$

$$11. y = (5x - 1)^4 \ln x$$

$$12. y = \frac{2x+5}{e^{9x}}$$

$$13. y = (x^2 + 3x + 1)e^x \ln x$$

$$14. y = \frac{5xe^x}{x+2}$$

$$15. y = \frac{6x-7}{x^2+1} - 5x \ln(2x - 5)$$